Leveraging Variational Autoencoders for Multiple Data Imputation

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Abstract. Missing data persists as a major barrier to data analysis across numerous applications. Recently, deep generative models have been used for imputation of missing data, motivated by their ability to learn complex and non-linear relationships. In this work, we investigate the ability of variational autoencoders (VAEs) to account for uncertainty in missing data through multiple imputation. We find that VAEs provide poor empirical coverage of missing data, with underestimation and overconfident imputations. To overcome this, we employ β-VAEs, which viewed from a generalized Bayes framework, provide robustness to model misspecification. Assigning a good value of β is critical for uncertainty calibration and we demonstrate how this can be achieved using cross-validation. We assess three alternative methods for sampling from the posterior distribution of missing values and apply the approach to transcriptomics datasets with various simulated missingness scenarios. Finally, we show that single imputation in transcriptomic data can cause false discoveries in downstream tasks and employing multiple imputation with β-VAEs can effectively mitigate these inaccuracies.

Keywords: VAEs · multiple imputation

1 Introduction

Missing data persists as a major barrier in analyses of multivariate data, due to issues like incomplete collection, data availability and low coverage. Early approaches for dealing with missing data tend to reduce the generalizability of results or skew the trends present in the data [38]. These include listwise deletion, where only complete observations are considered, or imputation methods, where the missing values are estimated. Some of these imputation strategies include substitution by the mean of observed values, stochastic regression techniques and hot deck imputation. Single imputation implicitly assumes that the imputation is perfect and thereby fails to account for the uncertainty introduced by the prediction. An attractive solution for this is multiple imputation, which

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models the uncertainty in the missing values by producing several plausible values for each imputed data point [30]. The imputed datasets are then combined and analyzed in downstream tasks to give estimates and standard errors that acknowledge uncertainty in the missing data.

Recently, deep generative models have emerged as a popular tool for imputing data, due to their ability to capture non-linear relationships and complex dependencies [3,7,10,12,17,21,23–26,29,31,32,34]. For example, Qiu et al. [34] use variational autoencoders (VAEs) for imputation of high-dimensional genomic data and find that it performs better than competing methods, such as singular value decomposition and K-nearest neighbours, but they focus solely on single imputation. In this work, we extend this approach to account for uncertainty through multiple imputation strategies. While expressive and powerful, deep models have been shown to be overconfident [39] and underestimate the variability of out-of-distribution test data [33]. In line with these results, we find that VAEs provide poor empirical coverage of the missing data, with underestimation and overconfident imputations for missing data values that are far from the mean.

To overcome this, we employ β-VAEs [14], which provide a framework for approximate Bayesian inference of deep generative models under the power likelihood. In statistics, inference based on the power likelihood has been shown to provide robustness against model misspecification [2], and thus, in our setting, it is crucial to avoid overfitting and achieve good coverage and well-calibrated uncertainty of the missing data. Assigning a good value of β is critical [15], and we employ cross-validation to tune β.

Lastly, we study the effects of imputation in downstream analyses. In the task of identifying discriminating gene sets in transcriptomic data from cancer cells, we show that single imputation can induce correlations that confound regression analyses. Further, we show that multiple imputation with β-VAEs yields fewer false positives in this task.

2 Background

2.1 Variational Autoencoders

Variational autoencoders [19] are probabilistic deep generative models comprised of two parts, the **encoder** and **decoder**. The encoder (also referred to as the inference model) takes an observed data point, $\mathbf{x} \in \mathbb{R}^D$, and computes the posterior distribution, $p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$, of the latent variables, $\mathbf{z} \in \mathbb{R}^K$. As the true posterior is intractable in most cases, an approximate model, $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, is used to approximate the true posterior, $p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$, and encode the observed data into the latent variables. The second part is the decoder (also referred to as the generative model) where the latent variables, $\mathbf{z}$, are used to reconstruct data point, $\hat{\mathbf{x}}$, via the generative model, $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$. The standard choice of distribution for both the inference and generative model is a simple, factorized Gaussian, where the Gaussian mean and variance are parameterized by neural networks, with $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$ containing the parameters of the encoder and decoder neural networks respectively. Based on a training data set $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_N)$ containing N data points, the neural network parameters $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$ are optimized during training of the VAE by minimizing the reconstruction loss (the negative log-likelihood of the data) and the latent loss (the Kullback-Leibler (KL) divergence between the variational posterior, $q_{\boldsymbol{\phi}}(\mathbf{Z}|\mathbf{X})$, and the prior (which we set to a standard Gaussian), $p(\mathbf{Z})$, with $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_N)$).

From a Bayesian perspective, this is equivalent to approximate variational inference of deep latent variable models, under the generative model $\mathbf{x}_n \mid \mathbf{z}_n \sim p_{\boldsymbol{\theta}}(\mathbf{x}_n \mid \mathbf{z}_n)$ with a Gaussian prior on latent variables $\mathbf{z}_n \sim \mathrm{N}(\mathbf{0}, \mathbf{I})$. To overcome the intractability of the posterior, amortized variational inference [11] is employed, assuming the variational posterior $q_{\boldsymbol{\phi}}(\mathbf{z}_n|\mathbf{x}_n)$ is parameterized by a neural network with $\boldsymbol{\phi}$ containing the weights and biases. The variational parameters $\boldsymbol{\phi}$ and generative model parameters $\boldsymbol{\theta}$ are optimized by minimizing the KL divergence between the variational posterior $q_{\boldsymbol{\phi}}(\mathbf{Z}|\mathbf{X})$ and the true posterior $p_{\boldsymbol{\theta}}(\mathbf{Z}|\mathbf{X})$, or equivalently maximizing the evidence lower bound (ELBO):

$$\text{ELBO} = \sum_{n=1}^{N} \mathbb{E}_{\mathbf{z}_n \sim q_{\phi}(\mathbf{z}_n|\mathbf{x}_n)}[\log p_{\boldsymbol{\theta}}(\mathbf{x}_n|\mathbf{z}_n)] - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}_n|\mathbf{x}_n), p(\mathbf{z}_n)).$$

During training, the ELBO is maximized using stochastic gradient descent. To compute the required gradients, the re-parameterization trick [18,35] is used to obtain independence between the latent noise and $\boldsymbol{\phi}$.

***β*-VAEs** An extension on the classic VAE is the β-VAE, which includes a hyperparameter β that adjusts the relative weight of the latent loss [14]:

$$\text{ELBO} = \sum_{n=1}^{N} \mathbb{E}_{\mathbf{z}_n \sim q_{\phi}(\mathbf{z}_n|\mathbf{x}_n)}[\log p_{\boldsymbol{\theta}}(\mathbf{x}_n|\mathbf{z}_n)] - \beta\, D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}_n|\mathbf{x}_n), p(\mathbf{z}_n)). \tag{1}$$

While in the machine learning community, β-VAEs are motivated by their improvement in disentangling the latent variables [5], we provide an alternative motivation from a statistical perspective. In particular, maximizing the β-VAE bound in (1), is equivalent to maximizing:

$$\sum_{n=1}^{N} \mathbb{E}_{\mathbf{z}_n \sim q_{\phi}(\mathbf{z}_n|\mathbf{x}_n)}[\log p_{\boldsymbol{\theta}}(\mathbf{x}_n|\mathbf{z}_n)^{1/\beta}] - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}_n|\mathbf{x}_n), p(\mathbf{z}_n)),$$

or minimizing the KL divergence between the variational posterior $q_{\boldsymbol{\phi}}(\mathbf{Z}|\mathbf{X})$ and the posterior under the power likelihood (see Appendix A.4 [36]):

$$p_{\boldsymbol{\theta},\beta}(\mathbf{Z}|\mathbf{X}) \propto \prod_{n=1}^{N} p_{\boldsymbol{\theta}}(\mathbf{x}_n|\mathbf{z}_n)^{1/\beta} p(\mathbf{z}_n).$$

The use of the power likelihood in Bayesian statistics provides frequentist guarantees of posterior consistency in nonparametric models [40], while the Bayesian model under the standard updating with $\beta = 1$ may be inconsistent [1]. Moreover, the power likelihood provides robustness to model misspecification [2]. Given the complex, high-dimensional nature of the deep generative model $p_{\boldsymbol{\theta}}(\mathbf{x}_n|\mathbf{z}_n)$, this acknowledges and allows for a mismatch between the generative model and the true data generating distribution.

2.2 Single Imputation with VAEs

VAEs are ideal for imputing missing transcriptomic data, as they can learn non-linear relationships and generate imputations over thousands of features. In this work, we develop the approach of Qiu et al. [34]. They first train the VAE using only the subset of complete data to optimize the parameters $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$. For each data point $n = 1, \ldots, N$, $\mathbf{x}_n$ can be split into two parts: $\mathbf{x}_{\text{obs},n}$ containing the observed features and $\mathbf{x}_{\text{mis},n}$ containing the missing features, where $\mathbf{X}_{\text{obs}} = (\mathbf{x}_{\text{obs},1}, \ldots, \mathbf{x}_{\text{obs},N})$ and $\mathbf{X}_{\text{mis}} = (\mathbf{x}_{\text{mis},1}, \ldots, \mathbf{x}_{\text{mis},N})$. For each data point with missing features, i.e. $\mathbf{x}_n \neq \mathbf{x}_{\text{obs},n}$, the optimal choice, under the squared error loss, is to impute with the mean under the generative model:

$$\begin{aligned}\widehat{\mathbf{x}}_{\text{mis},n} &= \mathbb{E}\left[\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}\right] = \int \mathbf{x}_{\text{mis},n}\, p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n})\, d\mathbf{x}_{\text{mis},n} \\ &= \int\int \mathbf{x}_{\text{mis},n}\, p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n}, \mathbf{z}_n \mid \mathbf{x}_{\text{obs},n})\, d\mathbf{z}_n d\mathbf{x}_{\text{mis},n}.\end{aligned}$$

This integral is intractable; thus in [34], it is approximated by iteratively computing: 1) the expectation of $\mathbf{z}_n$ (mean of the encoder) given $\widehat{\mathbf{x}}_{\text{mis},n}$ and $\mathbf{x}_{\text{obs},n}$:

$$\widehat{\mathbf{z}}_n = \int \mathbf{z}_n\, q_{\boldsymbol{\phi}}(\mathbf{z}_n \mid \widehat{\mathbf{x}}_{\text{mis},n}, \mathbf{x}_{\text{obs},n})\, d\mathbf{z}_n,$$

and 2) the expectation of $\widehat{\mathbf{x}}_{\text{mis},n}$ (mean of the decoder) given $\widehat{\mathbf{z}}_n$:

$$\widehat{\mathbf{x}}_{\text{mis},n} = \int \mathbf{x}_{\text{mis},n}\, p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}, \widehat{\mathbf{z}}_n)\, d\mathbf{x}_{\text{mis},n}.$$

Each missing value is initialized with the mean observed value for its respective feature, and subsequently re-estimated until convergence. Note that when the likelihood factorizes across features (e.g. factorized Gaussian), $p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}, \mathbf{z}_n) = p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n} \mid \mathbf{z}_n)$. In their paper, Qiu et al. [34] optimized the model and hyper-parameters through a grid search, claiming that the standard VAE ($\beta = 1$) and training for 250 epochs resulted in the lowest mean absolute error of the imputed values when compared to true values. However, the authors did not examine the uncertainty calibration nor did they consider multiple imputation.

2.3 Multiple Imputation

Multiple imputation (see e.g. [22,30,38]) improves upon single imputation by modelling the uncertainty associated with the imputed values. It does so by creating M plausible values for each missing data point. Combining the M datasets enables estimation of standard errors and confidence intervals for dataset derived statistics. Historically setting M between 3 and 5 was deemed adequate [37] others point to cases where it may be necessary to set M higher [13].

In multiple imputation we aim to obtain and simulate from the predictive distribution for the missing data given the observed data, i.e. $p(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}})$. In particular, we assume that $\mathbf{X}$ follows a distribution, $p(\mathbf{X}|\boldsymbol{\psi})$, where $\boldsymbol{\psi}$ is a collection of all parameters of the model. Then we can write our predictive distribution as:

$$p(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}}) = \int p(\mathbf{X}_{\text{mis}}, \boldsymbol{\psi}|\mathbf{X}_{\text{obs}})d\boldsymbol{\psi} = \int p(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}}, \boldsymbol{\psi})p(\boldsymbol{\psi}|\mathbf{X}_{\text{obs}})d\boldsymbol{\psi}.$$

To impute the missing data, and thereby simulate one of M plausible datasets, data augmentation (DA) algorithms can be employed. Specifically, DA is a Markov chain method which iteratively samples 1) the parameters $\boldsymbol{\psi}$ from the posterior $p(\boldsymbol{\psi}|\mathbf{X}_{\text{obs}}, \mathbf{X}_{\text{mis}})$ and 2) missing data $\mathbf{X}_{\text{mis}}$ given $\boldsymbol{\psi}$ from $p(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}}, \boldsymbol{\psi})$. This ultimately results in sampling from the predictive distribution $p(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}})$, producing one of the plausible datasets, denoted as $\mathbf{X}^m_{\text{mis}} = (\mathbf{x}^m_{\text{mis},1}, \ldots, \mathbf{x}^m_{\text{mis},N})$. This procedure is repeated M times to achieve M plausible datasets. Inferences based on these M imputed datasets can be combined via **Rubin's rules** to compute accurate inference about the entire dataset $\mathbf{X}$.

3 Methodology

In this work, we generalize single imputation with VAEs in two ways. First, we employ and compare three multiple imputation strategies to account for uncertainty in the missing data. Second, we extend using β-VAEs for improved robustness and uncertainty quantification. We propose an additional minor modification to the approach of Qiu et al., whereby we train the VAE by including both complete samples and those with some missingness. This requires that we apply mean imputation of missing values prior to training. We find this approach leads to modest improvements in imputation accuracy and allows the approach to be employed even in scenarios where the majority of samples contain some missingness.

3.1 Multiple Imputation with β-VAEs

In the case of multiple imputation (MI), the latent variables, $\mathbf{Z}$, of the β-VAE represent the parameters of our model, previously referred to as $\boldsymbol{\psi}$ in Sect. 2.3. To produce a sample from our target predictive distribution, $p_{\boldsymbol{\theta},\beta}(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}})$, we can iteratively sample from the joint distribution $p_{\boldsymbol{\theta},\beta}(\mathbf{X}_{\text{mis}}, \mathbf{Z}|\mathbf{X}_{\text{obs}})$ via a Markov chain Monte Carlo scheme. For β-VAEs, the predictive distribution is constructed from the power likelihood, that is the likelihood of our generative model is raised to the power $1/\beta$ (see Section 2.1):

$$\begin{aligned} p_{\boldsymbol{\theta},\beta}(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}}) &\propto \int p_{\boldsymbol{\theta}}(\mathbf{X}_{\text{mis}}|\mathbf{X}_{\text{obs}},\mathbf{Z})^{1/\beta} p_{\boldsymbol{\theta},\beta}(\mathbf{Z}|\mathbf{X}_{\text{obs}}) d\mathbf{Z} \\ &= \prod_{n=1}^{N} \int p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n}|\mathbf{x}_{\text{obs},n},\mathbf{z}_n)^{1/\beta} p_{\boldsymbol{\theta},\beta}(\mathbf{z}_n|\mathbf{x}_{\text{obs},n}) d\mathbf{z}_n, \end{aligned} \tag{2}$$

where standard VAEs correspond to $\beta = 1$. We note that in the case of the factored Gaussian generative model, the power likelihood $p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n}|\mathbf{x}_{\text{obs},n},\mathbf{z}_n)^{1/\beta}$ is simply proportional to a Gaussian with variance rescaled by a factor of β. On the other hand, $p_{\boldsymbol{\theta},\beta}(\mathbf{Z}|\mathbf{X}_{\text{obs}})$ represents the intractable true posterior of the latent variables under the power likelihood given the observed data only.

In the following, we implement and compare three different approaches to sample from our target predictive distribution in (2): 1) pseudo-Gibbs (Sect. 3.1), 2) Metropolis-within-Gibbs (Sect. 3.1), and 3) sampling importance resampling (Sect. 3.1). These strategies are proposed in [28,29,35], respectively, for missing data imputation with deep generative models, and we describe a simple extension based on β-VAEs and the power likelihood. Prior to imputation, we first train the β-VAE, using mean imputation for the missing values, to obtain estimates of generative model parameters $\boldsymbol{\theta}$ and variational parameters $\boldsymbol{\phi}$, and thus also an approximation of the true posterior of the latent variables.

Pseudo-Gibbs. Pseudo-Gibbs sampling was the first strategy developed to generate approximate samples from the predictive distribution in deep generative models [35]. In particular, approximate samples from the joint $p_{\boldsymbol{\theta},\beta}(\mathbf{X}_{\text{mis}},\mathbf{Z} \mid \mathbf{X}_{\text{obs}})$ are obtained by iteratively sampling from the encoder and decoder. More specifically, for $s = 1,\ldots,S$ iterations and every data point $n \in \{1,\ldots,N\}$ with missing features, the pseudo-Gibbs algorithm replaces the expectation steps in the single imputation of Sect. 2.2 with sampling, as follows: First, Sample $\mathbf{z}_n$ (sample of encoder) given $\mathbf{x}_{\text{mis},n}^{(s-1)}$:

$$\mathbf{z}_n^{(s)} \sim q_{\boldsymbol{\phi}}(\mathbf{z}_n \mid \mathbf{x}_{\text{mis},n}^{(s-1)}, \mathbf{x}_{\text{obs},n}). \tag{3}$$

Next, we sample $\mathbf{x}_{\text{mis},n}^{(s)}$ (sample of decoder) given $\mathbf{z}_n^{(s)}$ based on the power likelihood:

$$\mathbf{x}_{\text{mis},n}^{(s)} \sim p_{\boldsymbol{\theta},\beta}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}, \mathbf{z}_n^{(s)}) \propto p_{\boldsymbol{\theta}}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}, \mathbf{z}_n^{(s)})^{1/\beta}.$$

Ideally, in the first step, we would aim to sample from the intractable true posterior of the latent variables. However, if the variational posterior provides a good approximation, the pseudo-Gibbs scheme will produce samples from a distribution close to our target.

Metropolis-Within-Gibbs. The pseudo-Gibbs algorithm was improved and extended by [28], who derived a Metropolis-within-Gibbs (MWG) sampler that is asymptotically guaranteed to produce samples from the target predictive distribution. This is a simple modification of pseudo-Gibbs that corrects the first step by using the variational posterior as a proposal within a Metropolis-Hastings algorithm. Specifically, in the first step, the sampled value from the encoder in (3) represents the proposed value for the latent variables, denoted by $\mathbf{z}_n^*$, which is then accepted according to the acceptance probability:

$$a(\mathbf{z}_n^{(s-1)} \to \mathbf{z}_n^*) = \min\left(1, \frac{p_{\boldsymbol{\theta}}\left(\mathbf{x}_{\text{mis},n}^{(s-1)}, \mathbf{x}_{\text{obs},n} | \mathbf{z}_n^*\right)^{1/\beta} p(\mathbf{z}_n^*)}{p_{\boldsymbol{\theta}}\left(\mathbf{x}_{\text{mis},n}^{(s-1)}, \mathbf{x}_{\text{obs},n} | \mathbf{z}_n^{(s-1)}\right)^{1/\beta} p(\mathbf{z}^{(s-1)})} \frac{q_\phi\left(\mathbf{z}_n^{(s-1)} | \mathbf{x}_{\text{mis},n}^{(s-1)}, \mathbf{x}_{\text{obs}}\right)}{q_\phi\left(\mathbf{z}_n^* | \mathbf{x}_{\text{mis},n}^{(s-1)}, \mathbf{x}_{\text{obs},n}\right)}\right)$$

Thus, we set:

$$\mathbf{z}_n^{(s)} = \begin{cases} \mathbf{z}_n^* & \text{with prob. } a(\mathbf{z}_n^{(s-1)} \to \mathbf{z}_n^*) \\ \mathbf{z}_n^{(s-1)} & \text{with prob. } 1 - a(\mathbf{z}_n^{(s-1)} \to \mathbf{z}_n^*) \end{cases}.$$

If the variational posterior is a perfect approximation of the true posterior, the acceptance probability will be one, and the algorithm reduces to pseudo-Gibbs. In general, MWG acknowledges and corrects for the approximation of the posterior; however, if the variational posterior is far from the true posterior, MWG will suffer from low acceptance rates and slow convergence.

Sampling Importance Resampling. An alternative to Gibbs is sampling importance resampling (SIR), proposed by [29]. First, we perform importance sampling using the variational posterior as the importance distribution. In this case, for every data point $n \in \{1, \ldots, N\}$ with missing features, we take $s = 1, \ldots, S$ samples of the latent variables from our importance distribution.

$$\mathbf{z}_n^{(s)} \sim q_\phi(\mathbf{z}_n \mid \mathbf{x}_{\text{mis},n}^{(0)}, \mathbf{x}_{\text{obs},n}),$$

where $\mathbf{x}_{\text{mis},n}^{(0)}$ denotes an initial mean imputation for the missing data (zero is the mean after feature standardization). These importance samples $(\mathbf{z}_n^{(s)})$, for $s = 1, \ldots, S$, have weights $w_n^{(s)}$ proportional to:

$$\omega_n^{(s)} = \frac{p_{\boldsymbol{\theta}}\left(\mathbf{x}_{\text{obs},n} | \mathbf{z}_n^{(s)}\right)^{1/\beta} p(\mathbf{z}_n^{(s)})}{q_\phi\left(\mathbf{z}_n^{(s)} | \mathbf{x}_{\text{mis},n}^{(0)}, \mathbf{x}_{\text{obs},n}\right)},$$

where $w_n^{(s)} = \omega_n^{(s)} / \sum_{s=1}^S \omega_n^{(s)}$ (for further details, see Appendix A.5 [36]). Then, for multiple imputation, we obtain M imputations by first sampling $(\mathbf{z}_n^m)$, for $m = 1, \ldots M$, with replacement from the importance samples $(\mathbf{z}_n^{(s)})$ with probability $w_n^{(s)}$. Next, for each $\mathbf{z}_n^m$, we impute the missing data by sampling from

$$\mathbf{x}_{\text{mis},n}^m \sim p_{\boldsymbol{\theta},\beta}(\mathbf{x}_{\text{mis},n} \mid \mathbf{x}_{\text{obs},n}, \mathbf{z}_n^m).$$

In contrast to Gibbs sampling, an advantage of SIR is parallelizability. However, the discrepancy between the variational posterior and true posterior determines the efficiency of the algorithm, and a large discrepancy may result in degeneracy of the weights and require a large number of importance samples (which is required to be exponential in KL divergence between the importance distribution and the target [4]).

3.2 Cross-Validation Training Regime

When the generative model and $\boldsymbol{\theta}$ match the true data generating distribution exactly, learning is achieved optimally with $\beta = 1$. However, in practice, we have a mismatch and assigning a good value of β becomes critical to achieve robustness and accurate uncertainty quantification. Indeed, if β is set too low, the posterior uncertainty can be underestimated, while if β is set too high, the posterior uncertainty is overestimated. Some directions for assigning a value of β from an information theoretic perspective are provided in [15]. Instead, we employ cross-validation (CV) to tune β for accurate multiple imputation and coverage of the missing data. A second consideration that affects uncertainty calibration in deep models is overfitting. We observe that training for too long leads to under-estimated uncertainty on the held-out data (Supplementary Figure A.3 [36]). Therefore, selecting the correct number of epochs is critical for well-calibrated uncertainty. This motivates the CV approach to select β and the number of epochs jointly.

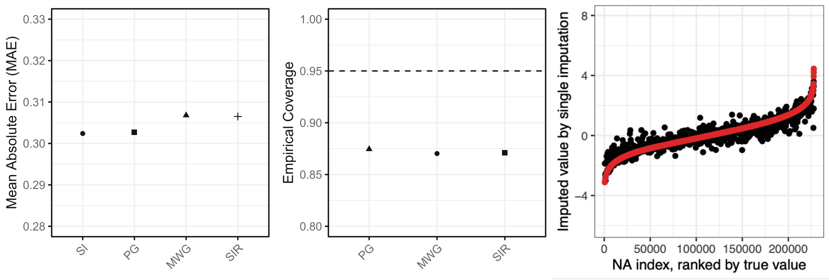
Specifically, the CV approach to tuning β and the number of epochs consists of creating k copies of the data and adding a small proportion of additional MCAR missingness in each copy. We then carry out a grid search over the number of epochs and values of β, training k models for each value of β. The final selection is the combination that has acceptable coverage while minimizing the mean absolute error (MAE) over the introduced missing values (averaged across the k models). Optimal values are selected by visual inspection of the MAE and coverage in the CV plots (Supplementary Figure A.3 [36]). Once the optimal hyper-parameters for β and epochs are selected, the model is retrained using all of the data. We observed that following this approach results in coverage and MAE on the test set being close to the values estimated through cross-validation.

3.3 Evaluating Imputation Performance

To evaluate the imputation performance, we consider two quantities: 1) the mean absolute error (MAE) to assess reconstruction accuracy and 2) the empirical coverage (EC) to quantify uncertainty. The MAE compares our imputed values to the ground truth that was originally masked in the complete dataset. Recall that $\widehat{\mathbf{X}}_{\text{mis}} = (\hat{\mathbf{x}}_{\text{mis},1}, \ldots, \hat{\mathbf{x}}_{\text{mis},N})$ represents the imputed values, while $\mathbf{X}_{\text{mis}}$ represents the true (masked) values. The MAE is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^{N} |\widehat{\mathbf{x}}_{\text{mis},n} - \mathbf{x}_{\text{mis},n}|, \tag{4}$$

where $|\widehat{\mathbf{x}}_{\text{mis},n} - \mathbf{x}_{\text{mis},n}|$ represents the average absolute difference across all missing features for the nth data point. For multiple imputation, the imputed values



(a) Accuracy (b) Coverage (c) Imputed values by SI

Fig. 1. Standard VAE (β=1) underestimates the uncertainty in multiple imputation. Here we report **(a)** the accuracy at imputed missing values compared to the ground truth by MAE and **(b)** the fraction of true values that fall within the 95% CIs for the three multiple imputation approaches pseudo-Gibbs (PG), Metropolis-within-Gibbs (MWG) and sampling importance resampling (SIR). Dotted line represents the desired coverage at 0.95. Finally, **(c)** depicts the imputed values for the missing data by single imputation (SI), ranked by their true values (highlighted in red). (Color figure online)

in (4) are averaged across the M imputed datasets, $\widehat{\mathbf{x}}_{\text{mis},n} = \frac{1}{M}\sum_{m=1}^{M}\mathbf{x}^{m}_{\text{mis},n}$. To evaluate uncertainty in multiple imputation, we first compute $100(1-\alpha)\%$ confidence intervals (CIs) for each missing value based on the M imputed values. The empirical coverage is then computed as the fraction of times where the true value falls within the predicted interval.

4 Results: Genomic Data Imputation

4.1 Limitations of Single Imputation and Standard VAEs

We compare single imputation with standard VAEs to multiple imputation using three methods to sample the posterior (MWG, PG, and SIR). In order to benchmark against [34] we employed the same RNA-sequencing dataset from the Cancer Genome Atlas (TCGA). This dataset contains $D = 17{,}175$ features (RNA-sequencing counts) for $N = 667$ glioma patients, comprised of two cancer subtypes, glioblastoma (GBM) and low-grade glioma (LGG). We first simulate missingness in this dataset by masking 10% of values completely at random (MCAR) in 20% of samples, scale the dataset and subsequently train the VAE with the zero imputation at missing value indices (see Sect. 3). In order to benchmark against their method, we use the same model and hyper-parameters that were found to be optimal in [34], specifically, the standard VAE (β=1) with 250 training epochs and a learning rate of 10^{-5}. Once our model is trained, we generate $M = 100$ plausible datasets for each multiple imputation approach and perform single imputation (as described in Sect. 2.2). We set M much higher than would typically be used in MI, in order to measure empirical coverage with a high degree of precision. We examine the effect of varying M as a hyper-parameter in our analysis (Supplementary Figure A.1 [36]).

To evaluate imputation of the original masked values, we consider imputation accuracy by MAE and find that the multiple imputation approaches have similar accuracy to single imputation (SI), with pseudo-Gibbs performing slightly better than the other MI approaches (Fig. 1a). Next, we consider the empirical coverage of the masked values based on the 95% CIs computed from the $M = 100$ imputed datasets for all three multiple imputation approaches, and find that the uncertainty is underestimated when $\beta = 1$ (Fig. 1b). Additionally, the values imputed at masked data points with single imputation are underestimated at more extreme true values (Fig. 1c). As [34] only used reconstruction accuracy with single imputation to optimize hyperparameters, they were unable to assess uncertainty calibration in the imputations. To overcome this, we explore regularization of the latent space through β-VAEs, optimizing the hyperparameters by considering both reconstruction accuracy and coverage.

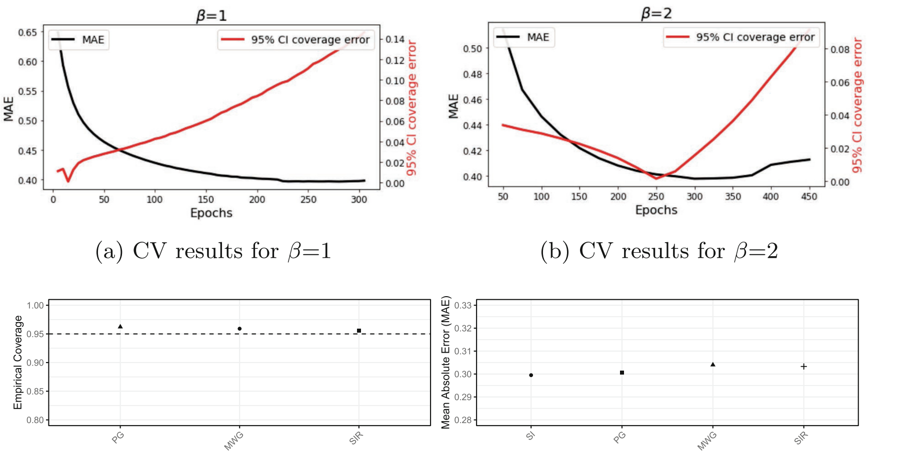
4.2 Multiple Imputation with β-VAEs for Accurate Uncertainty Quantification

For improved robustness, we employ β-VAEs and the cross-validation scheme described in Sect. 3.2 to tune β and the number of training epochs, resulting in a value of $\beta = 2$ and 250 training epochs (Supplementary Figure A.3 [36]). We then train the β-VAE with these optimal parameters and impute values by SI and all three MI approaches PG, MWG and SIR. This results in good coverage at 95%, with a much lower deviation from the desired coverage than the standard VAE with $\beta = 1$ (Fig. 2, Supplementary Figure A.2, Supplementary Figure A.5, Supplementary Figure A.6 [36]). Even with regularization of the latent space, single imputation still results in underestimation at extreme values (Supplementary Figure A.2 [36]). Our multiple imputation by β-VAEs yields good coverage across all missing data, even extreme values, while still retaining comparable accuracy to single imputation (Fig. 2d).

4.3 Multiple Imputation Reduces False Positives in Downstream Tasks

We next investigate the impact of all imputation approaches on downstream tasks, namely in identifying discriminating gene sets through logistic regression with the LASSO penalty. In particular, we run LASSO regression on all imputed datasets to identify the genes which discriminate between the two cancer subtypes, GBM and LGG. This results in one gene set from our ground truth dataset with no missingness (GT), one from single imputation (SI), and 100 discriminating gene sets for each multiple imputation approach, PG, MWG and SIR. We propose that non-zero LASSO coefficients that arise in the imputed data but do not show up in the complete data LASSO model are false positives arising as artefacts of imputation.

We find that the union across all discriminating gene sets for each multiple imputation approach is much larger than the ground truth set, with the total number of possible non-zero coefficients ranging from 143 to 155, and only 31



(a) CV results for β=1

(b) CV results for β=2

(c) 95% Coverage for $\beta = 2$

(d) Accuracy for $\beta = 2$

Fig. 2. Multiple imputation with β-VAEs provides calibrated coverage. The standard VAE (β=1) cannot achieve a balance of high accuracy and calibrated coverage **(a)**. While the β-VAE with β=2 **(b)** gets close to the error minimum while maintaining accurate uncertainty estimation (as measured by empirical coverage with a 95% CI. After imputing by SI, MWG, PG and SIR, we summarize imputation performance by **(c)** the empirical coverage at 95% CIs (dotted line represents the desired coverage at 0.95) and **(d)** the accuracy at imputed values, comparing single imputation with all three MI strategies.

discriminating genes in the true dataset (Table 1). When comparing the estimated coefficients from the ground truth data to the (averaged) estimated coefficients based on the imputed data, this results in a slightly higher MAE for multiple imputation approaches (0.066, 0.064 and 0.069 for PG, MWG and SIR, respectively) compared to single imputation (0.053), which also has a set of 31 discriminating genes, although these are not identical to the ground truth set. However, when we inspect the coverage across the multiple imputations, we find that our coverage is close to the desired 95% across PG, MWG and SIR (Table 1, Supplementary Figure A.7 [36]).

To identify discriminating gene sets across multiple imputations, we consider two approaches: selecting genes that 1) do not include a coefficient of zero in the 95% CI computed from the 100 imputed datasets, and 2) have an inclusion probability, denoted P_{incl} and defined as the fraction of imputed datasets that the gene has a non-zero LASSO coefficient, greater than a specified threshold. The first approach results in the same set of 12 genes across all three multiple imputation approaches that are all in the true set of non-zero LASSO coefficients (Table 1, Fig. 3), giving a false discovery rate (FDR) of 0%. These 12 genes are also contained within the set for single imputation; however, single imputation results in 7 false positives (Fig. 3), yielding an FDR of 22.6% (7/31). In the second approach, if we threshold at $P_{\text{incl}} > 0.5$, this results in a final set of 25

Table 1. Performance of different imputation techniques, single imputation (SI) and multiple imputation by PG, MWG and SIR for imputation at missing value indices (first two rows) and downstream impact on LASSO regression (subsequent rows). The final row reports the false discovery rate, based genes with an inclusion probability > 0.5 for multiple imputation.

Metric	SI	PG	MWG	SIR
MAE	0.302	0.301	0.304	0.303
95% CI coverage	N/A	96.2%	95.9%	95.6%
LASSO: MAE	0.053	0.066	0.064	0.069
LASSO: 95% CI coverage	N/A	97.4%	97.2%	96.6%
LASSO: total number of non-zero coefficients	31	155	143	149
LASSO: number of genes without zero in 95% CI	N/A	12	12	12
LASSO: number of genes with $P_{\text{incl}} > 0.5$	N/A	25	25	25
LASSO: False discovery rate	22.6%	8%	8%	8%

discriminating genes for each multiple imputation approach (Table 1, Supplementary Figure A.8 [36]). In this case, our gene set contains 2 false positives, yielding an FDR of 8.0% (Fig. 3). In summary, we find that multiple imputation with β-VAEs not only provides well-calibrated uncertainty but also results in much more acceptable FDRs in downstream tasks.

4.4 Multiple Imputation in New Missing Scenarios and Additional Transcriptomic Datasets

Lastly, we show that our results are consistent when applied to another transcriptomic dataset and with different missingness scenarios. As missing-not-at-random (MNAR) is common in transcriptomic data, due to either low coverage or artefacts of the sequencing protocol, we choose to explore this additional missing scenario [6]. Here we select genes with GC content at the highest 10% and randomly mask half of these values to generate our MNAR simulated dataset. We additionally explore a new pan-cancer transcriptomic dataset, which is comprised of 17,175 features and 953 samples across 2 cancer types, lung adenocarcinoma (LUAD) and lung squamous carcinoma (LUSC). This makes up four groups for which we run our imputation framework: **(1)** MCAR for LGG and GBM, **(2)** MNAR for LGG and GBM, **(3)** MCAR for LUAD and LUSC and **(4)** MNAR for LUAD and LUSC.

For all four groups, we simulate missingness as described previously. We first tune β and the number of epochs by simulating additional MCAR missingness, then train the VAE with these tuned hyper-parameters before imputing at missing values. We evaluate accuracy by MAE. Coverage is evaluated by looking at the EC and the FPR in the downstream task of identifying discriminating gene sets with LASSO regression. We find that running cross-validation to tune β on the new transcriptomic dataset (LUAD-LUSC) results in the selection of

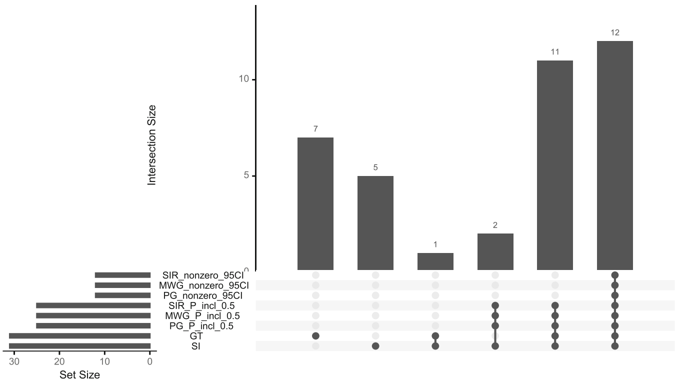
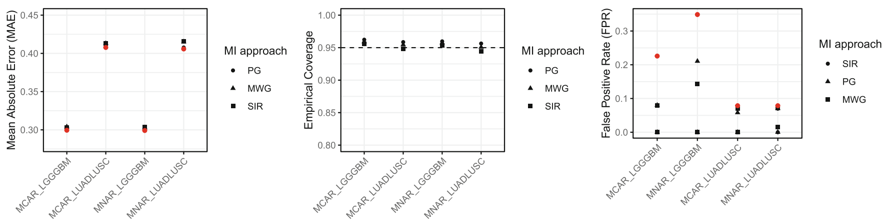


Fig. 3. Upset plot of discriminating gene sets from different imputation approaches. We report the discriminating gene sets by single imputation (SI), ground truth (GT), and all three multiple imputation approaches PG, MWG and SIR with two different inclusion criteria, zero not contained in 95% CI (nonzero_95CI) and inclusion probability, $P_{\text{incl}} > 0.5$ (P_incl_0.5).



(a) MAE (b) EC (c) FPR

Fig. 4. Multiple imputation across new missing scenarios and new transcriptomic datasets. Across different missing scenarios (MCAR and MNAR) for both transcriptomic datasets (LGG versus GBM and LUAD versus LUSC), we report the **(a)** MAE, **(b)** EC (empirical coverage), and **(c)** false positive rate (FPR) of discriminating gene sets from LASSO regression defined by SI and $P_{incl} = 0.5$ for MI settings. In plots **(a)** and **(c)**, the red dot represents the estimate from the single imputation approach.

approximately the same optimal values for β and the number of epochs ($\beta = 2$ and $epochs = 250$). Our results show that in all four contexts, we are able to retain imputation accuracy compared to single imputation (Fig. 4a) while still estimating coverage appropriately at the 95% level (Fig. 4b). Evaluating the discriminating gene sets identified by LASSO regression in the new LUAD-LUSC dataset and MNAR missingness scenarios, we repeat our finding that the false positive rate (Fig. 4c) is reduced and precision is increased (Supplementary Figure A.9 [36]) across all MI settings compared to SI. Here we have used the gene set for $P_{\text{incl}} > 0.5$ for each MI setting to increase our true positive rate, but we see this holds for the more stringent set as well, which are genes that do not include a coefficient of zero in the 95% CI computed from the 100 imputed datasets (Supplementary Figure A.10 [36]).

5 Discussion

We describe a deep learning framework for multiple imputation using β-VAEs. We propose and compare three multiple imputation methods and develop a new training regime, which uses all observed data to tune hyperparameters by assessing accuracy as well as empirical coverage. Our approach captures the complex, non-linear relationships present in high-dimensional genomic data, imputing values with high accuracy while retaining good coverage. Previous work [34] employed standard VAEs for genomic data imputation by single imputation, resulting in inaccurate and overconfident imputations at extreme missing values. More recent work has investigated multiple imputation using VAEs, but has not leveraged the potential of tuning β for proper uncertainty calibration [27]. Here we investigate the impact of these different imputation approaches on downstream tasks, namely discriminating gene sets identified by logistic regression with the LASSO penalty. We find that multiple imputation through β-VAEs identifies genes that discriminate between the two cancer subtypes with lower false discovery rates than previous methods. All three multiple imputation approaches perform similarly in terms of accuracy and coverage though SIR may be preferred as it permits parallelization (Supplementary Figure A.11 [36]).

Future work will continue to investigate missing not at random settings [7,17] and mixed data [24]. In addition, extensions using ensembles of deep generative models may improve robustness and calibration. Such ensembles can be built from simple approaches, such as training with multiple initializations [20], composing models across different epochs [16], or Monte Carlo dropout [9], to more advanced approaches, such as Bayesian methods [8].

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Code is available at https://github.com/roskamsh/BetaVAEMultImpute along with the appendix for this paper, including supplementary figures.

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